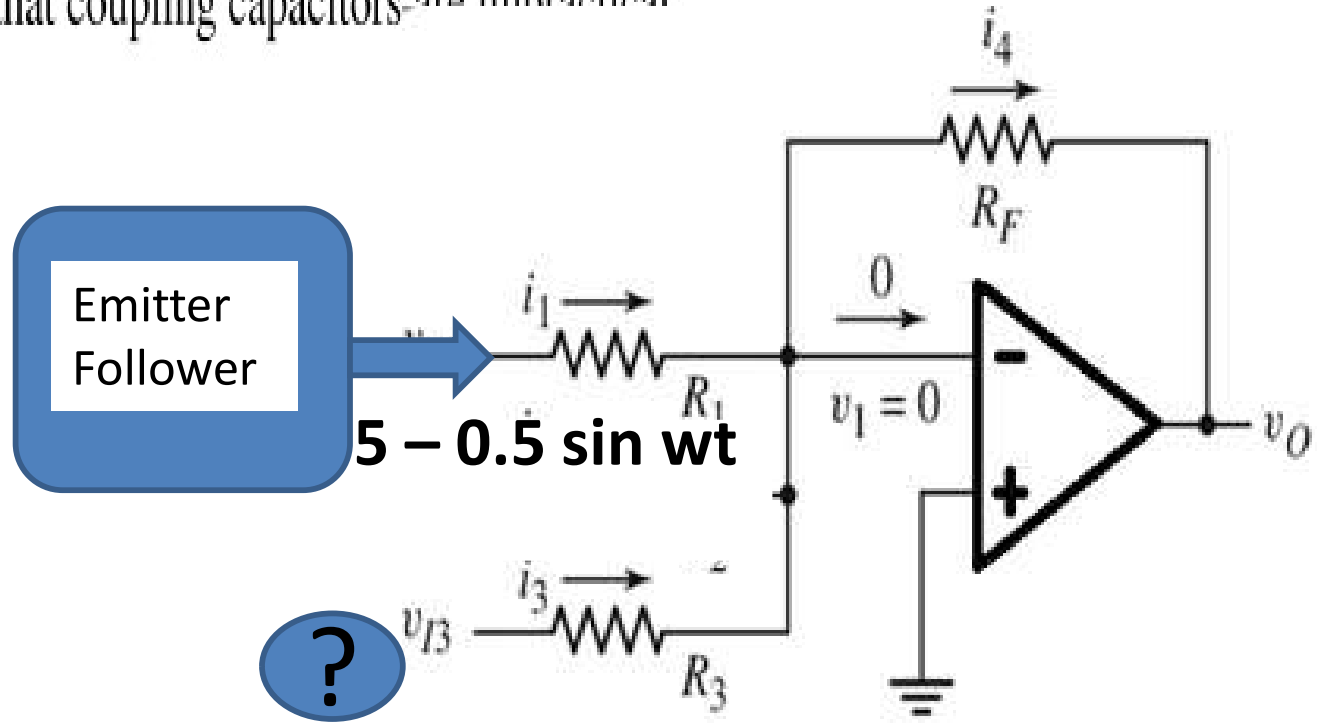
Design summing Amplifier

Objective: Design a summing amplifier to produce a specified output signal.

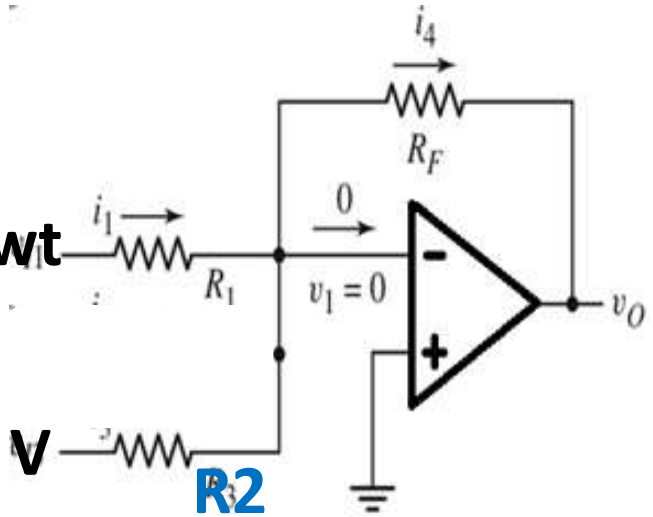
Specifications: The output signal generated from a BJT emitter follower amplifier is $v_{O1} = 5 - 0.5 \sin \omega t$ (V) and the effective output resistance is $R_o = 50\ \Omega$. Design a summing amplifier to be connected to the emitter follower such that the output signal is $v_O = 2 \sin\ \omega t$ (V). Assume that the frequency ω is quite low, such that coupling capacitors are impractical



Second input should be DC input to cancel 5 V from emitter follower output.

Let us keep same gain for both inputs and apply -5 V to second input to cancel +5V.

5 – 0.5 sin wt



Vo = 2 sin wt

R1 = R2 = 30 KΩ

Gain needed is -4

So, Rf / R1 = 4

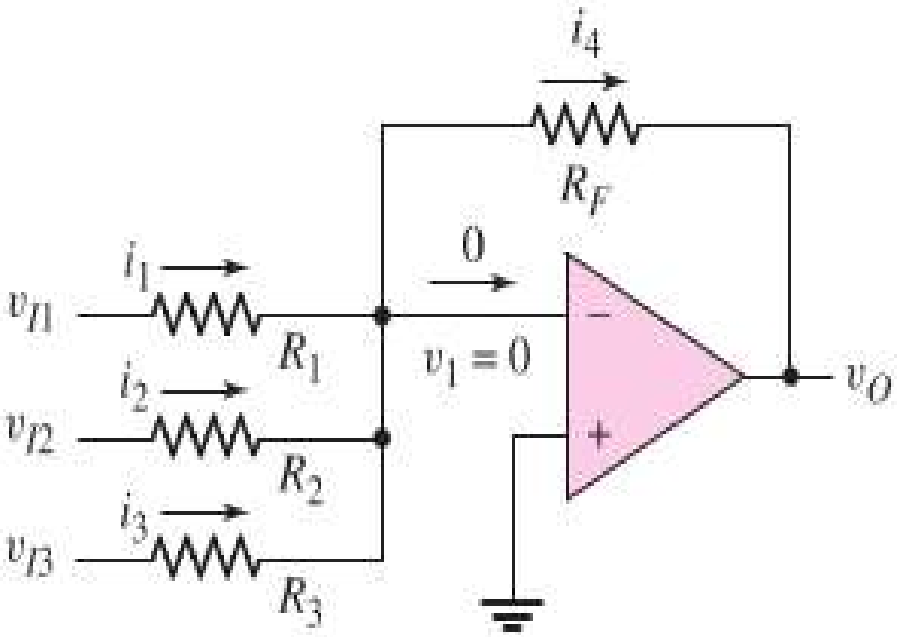
Rf = 120 KΩ

Ex 9.4: Design a summing amplifier that will produce an output voltage of $v_O = -(7v_{I1} + 14v_{I2} + 3.5v_{I3} + 10v_{I4})$. The maximum allowable resistance value is 280 kΩ.

$R_F = 280\ \text{k}\Omega$, $R_1 = 40\ \text{k}\Omega$, $R_2 = 20\ \text{k}\Omega$, $R_3 = 80\ \text{k}\Omega$, and $R_4 = 28\ \text{k}\Omega$

Design summing amplifier to produce average of three input voltages. Vo = (V1 + V2+ V3)/3

Design such that each input signal sees maximum possible input resistance under the condition that maximum allowed resistance in the circuit is 1M Ω.



R1 = R2 = R3 = 1 M

Rf = ?

333K

Determine output voltage Vo.

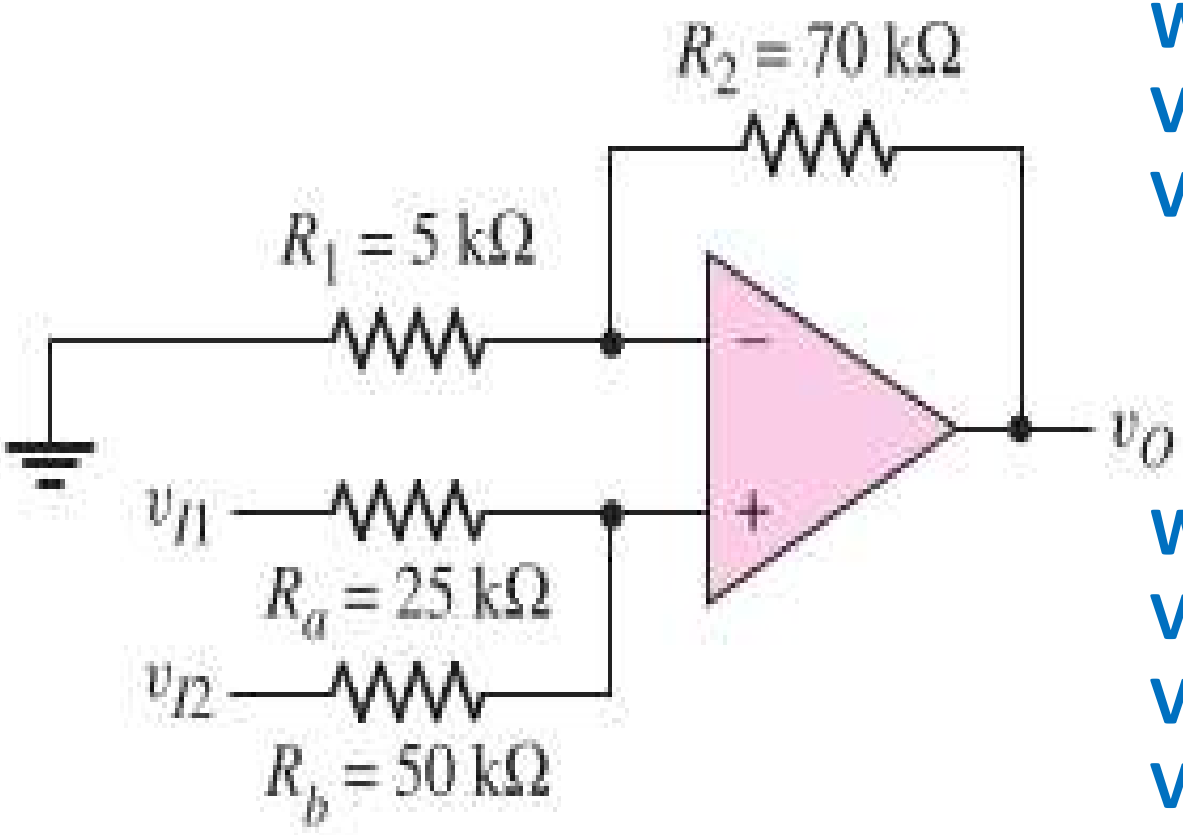
$$\underset{A\to\infty}{G} = 1 + \frac{R2}{R1}$$

G =15

When Vi2=0

Vp= Vi1[50/(25+50)] = 2/3Vi1

Vo1 = Vi1[2/3 *15] =10Vi1



When Vi1=0

Vp=Vi2[25/(25+50)] = 1/3 Vi2

Vo1 = Vi2[1/3 *15] =5Vi2

$$v_O = 10v_{I1} + 5v_{I2}$$

V to I converter

- We may want to drive coil in magnetic circuit , with given current , using voltage source.

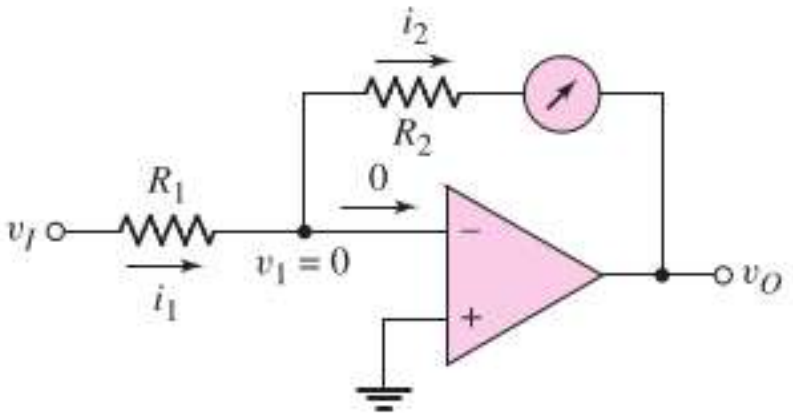


Figure 9.21 Simple voltage-to-current converter

$$i_2 = i_1 = V_I/R_1$$

Not Practical

As this means i_2 is directly proportional to V_I and not dependent on R_2

Is inverting terminal at virtual ground?

- $V_1 = V_2 = V_L = I_L Z_L$
- $i_1 = i_2$

$$\frac{v_I - i_L Z_L}{R_1} = \frac{i_L Z_L - v_O}{R_F} \quad \text{------1}$$

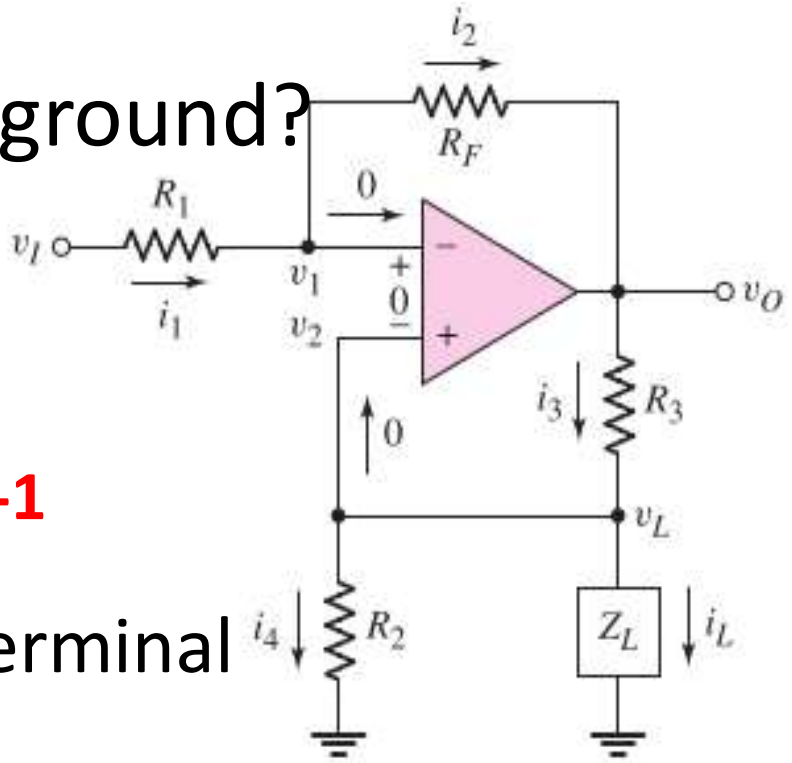


Figure 9.22 Voltage-to-current converter

Summing current at noninverting terminal

$$\frac{v_O - i_L Z_L}{R_3} = i_L + \frac{i_L Z_L}{R_2} \quad \text{------2}$$

- Solving for ($V_o - I_L Z_L$) from 1 and substituting in 2

$$\frac{R_F}{R_1} \cdot \frac{(i_L Z_L - v_I)}{R_3} = i_L + \frac{i_L Z_L}{R_2}$$

$$\frac{R_F}{R_1} \cdot \frac{(i_L Z_L - v_I)}{R_3} = i_L + \frac{i_L Z_L}{R_2}$$

Combining terms in i_L, we obtain

$$i_L \left(\frac{R_F Z_L}{R_1 R_3} - 1 - \frac{Z_L}{R_2} \right) = v_I \left(\frac{R_F}{R_1 R_3} \right) \quad \text{------3}$$

To make i_L independent of Z_L, make coefficient of Z_L as zero or

$$\frac{R_F}{R_1 R_3} = \frac{1}{R_2}$$

Equation 3 becomes

$$i_L = -v_I \left(\frac{R_F}{R_1 R_3} \right) = \frac{-v_I}{R_2}$$

$$i_L = -v_I \left(\frac{R_F}{R_1 R_3} \right) = \frac{-v_I}{R_2}$$

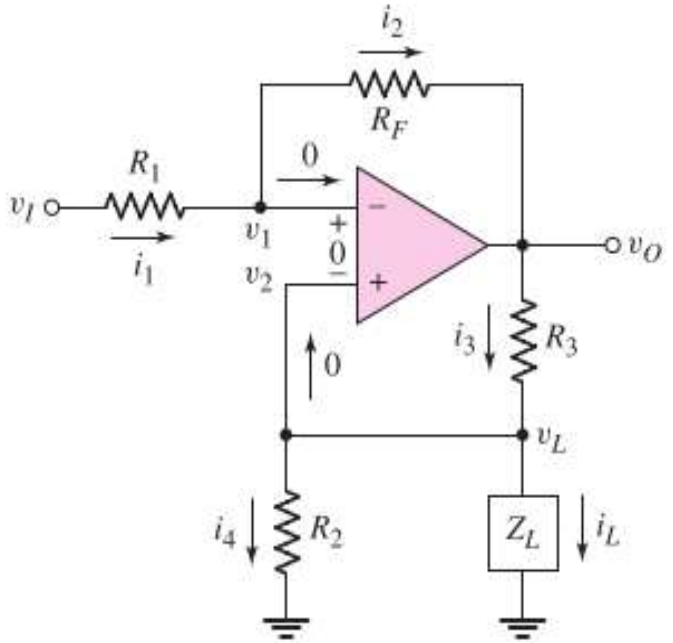


Figure 9.22 Voltage-to-current converter

which means that the load current is proportional to the input voltage and is independent of the load impedance Z_L, as long as the output voltage remains between allowed limits.

We may note that the input resistance seen by the source v_I is finite, and is actually a function of the load impedance Z_L. For a constant i_L, a change in Z_L produces a change in $v_L = v_2 = v_1$, which causes a change in i_1. A voltage follower may be inserted between the voltage source v_I and the resistor R_1 to eliminate any loading effects due to a variable input resistance.

EXAMPLE 9.5

Objective: Determine a load current in a voltage-to-current converter.

Consider the circuit in Figure 9.22. Let $Z_L = 100\,\Omega$, $R_1 = 10\,\text{k}\Omega$, $R_2 = 1\,\text{k}\Omega$, $R_3 = 1\,\text{k}\Omega$, and $R_F = 10\,\text{k}\Omega$. If $v_I = -5$ V, determine the load current i_L and the output voltage v_O.

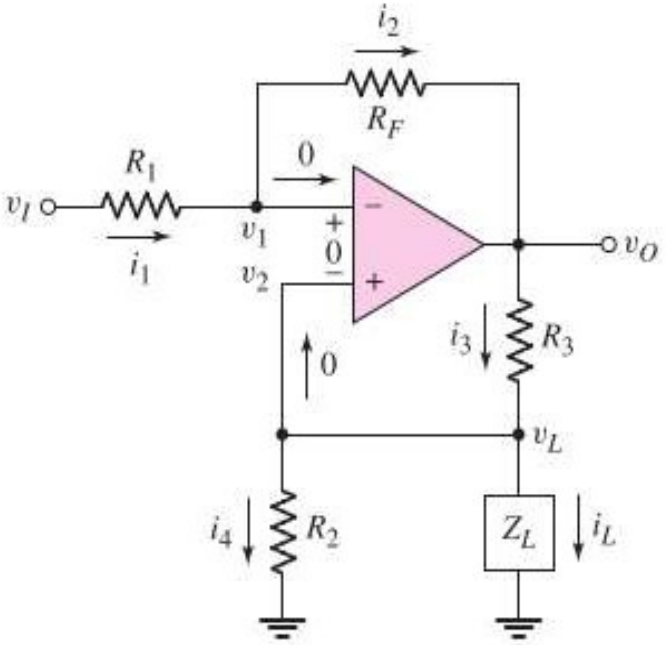


Figure 9.22 Voltage-to-current converter

Check: To make i_L independent of Z_L, make coefficient of Z_L as zero

$$\frac{1}{R_2} = \frac{R_F}{R_1 R_3} = \frac{10}{(10)(1)} \rightarrow \frac{1}{1}$$

The load current is

$$i_L = \frac{-v_I}{R_2} = \frac{-(-5)}{1\,\text{k}\Omega} = 5\,\text{mA}$$

and the voltage across the load is

$$v_L = i_L Z_L = (5 \times 10^{-3})(100) = 0.5\,\text{V}$$

What are values of i3 & i4 ?

$$i_4 = \frac{v_L}{R_2} = \frac{0.5}{1} = 0.5\,\text{mA}$$

and

$$i_3 = i_4 + i_L = 0.5 + 5 = 5.5\,\text{m}$$

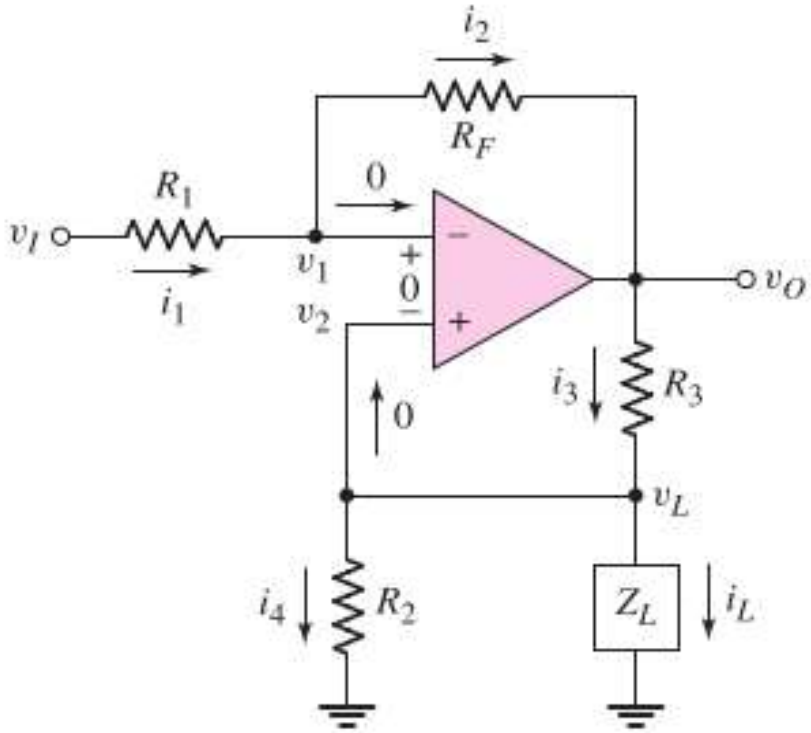


Figure 9.22 Voltage-to-current converter

The output voltage is then

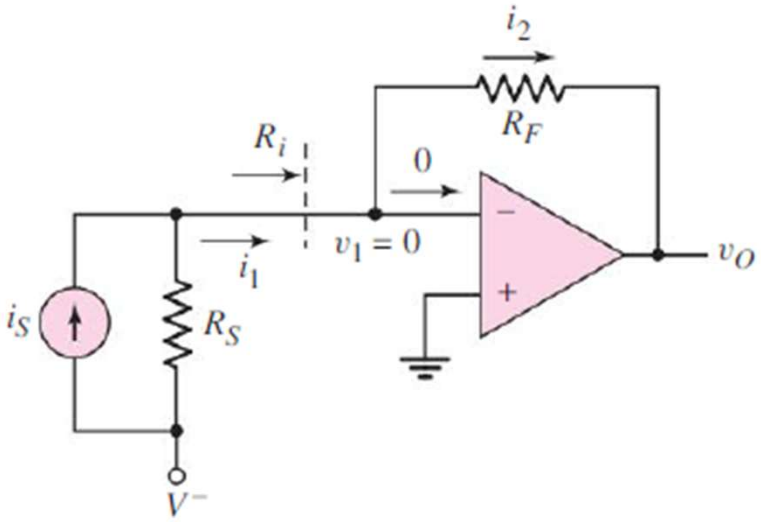
$$v_O = i_3 R_3 + v_L = (5.5 \times 10^{-3})(10^3) + 0.5 = 6\,\text{V}$$

We could also calculate i_1 and i_2 as

$$i_1 = i_2 = -0.55\,\text{mA}$$

Comment: In this example, we implicitly assume that the op-amp is not in saturation, which means that the applied dc bias voltage must be greater than 6 V. In addition, since currents i_2 (which is negative) and i_3 must be supplied by the op-amp, we are assuming that the op-amp is capable of supplying 6.05 mA.

Current-to-Voltage Converter



The input resistance Ri at the virtual ground node is

$Ri = v_1 / i_1 \sim= 0$

In most cases, assume that $R_S >> Ri$; therefore, current i1 is essentially equal to the signal current i_S .

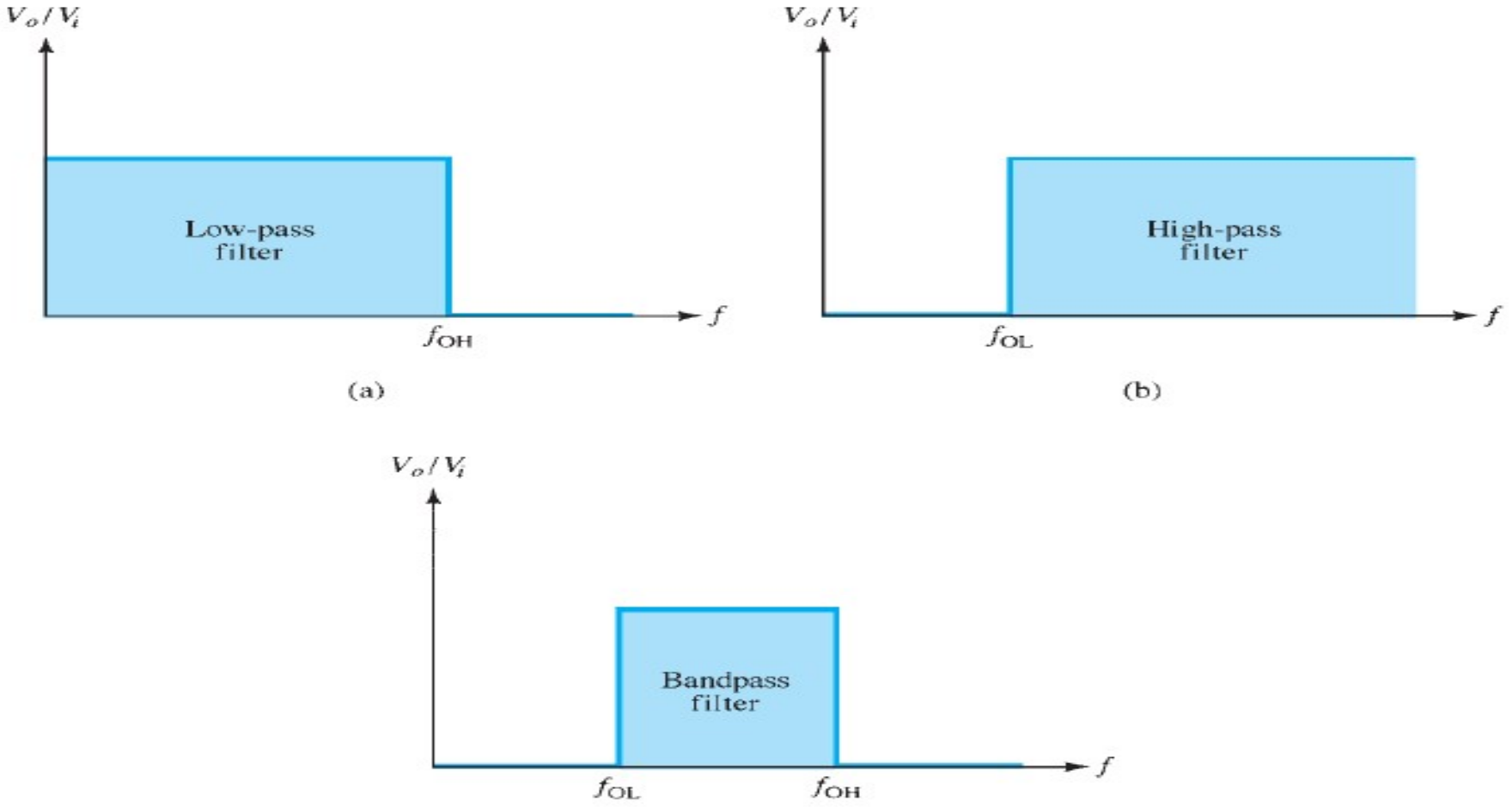
$i_2 = i_1 = i_S$

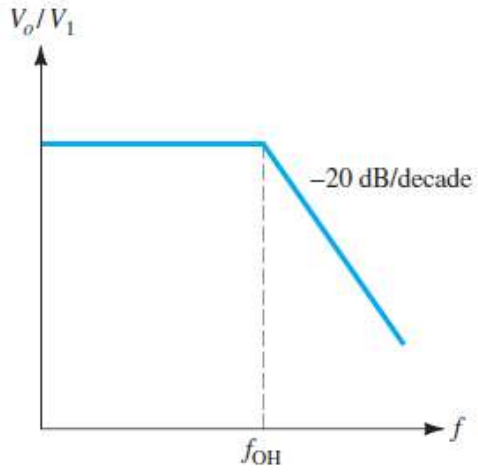
and

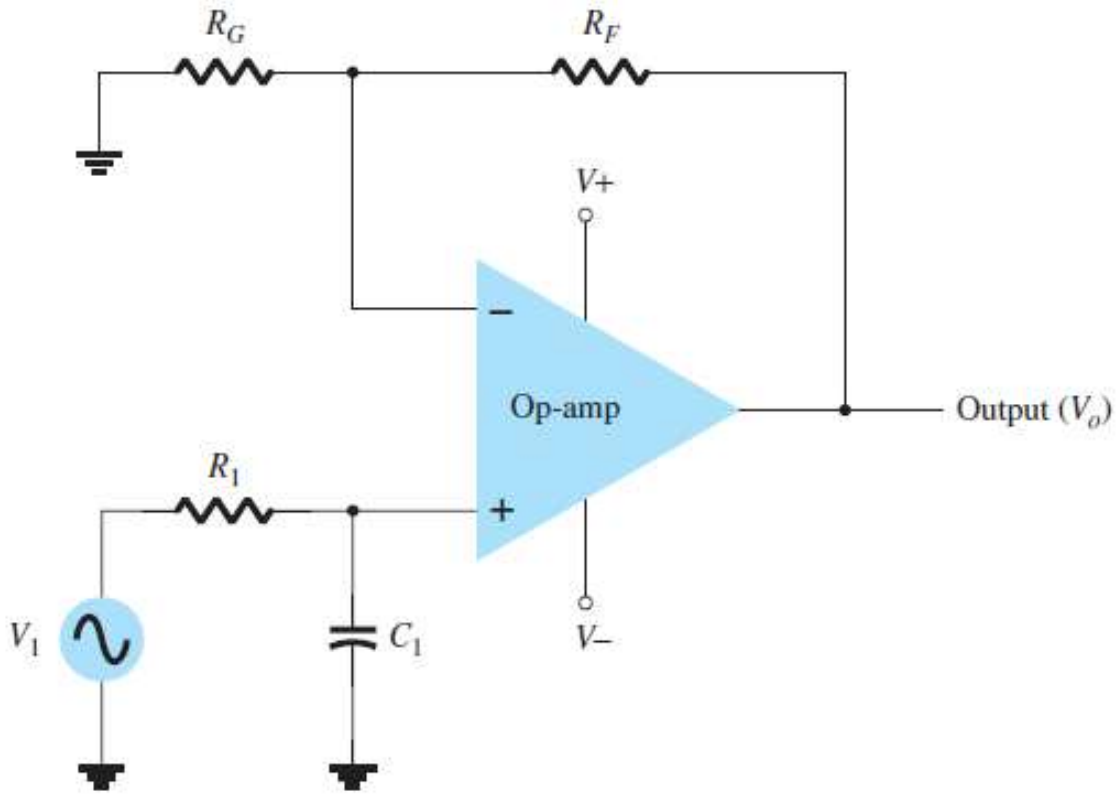
$V_O = - i_2 R_F = -i_S R_F$

The output voltage is directly proportional to the signal current, and the feedback resistance R_F is the magnitude of the ratio of the output voltage to the signal current

Active Filter





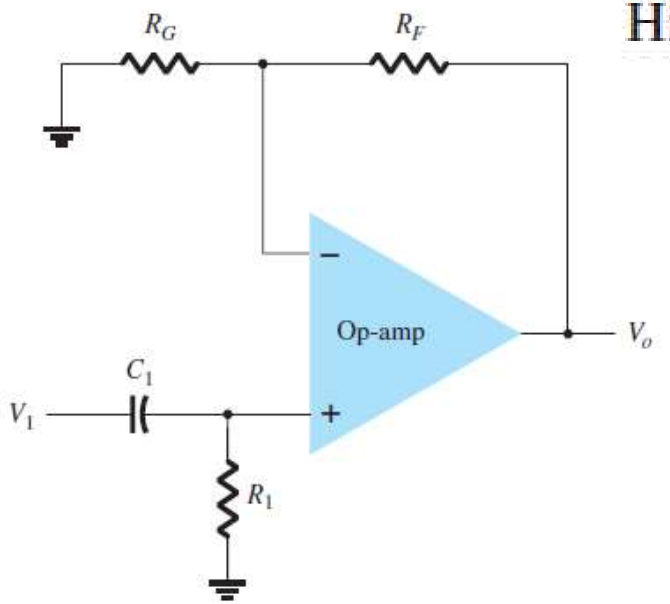


EXAMPLE 11.12 Calculate the cutoff frequency of a first-order low-pass filter for $R_1 = 1.2\ \text{k}\Omega$ and $C_1 = 0.02\ \mu\text{F}$.

Solution:

$$f_{OH} = \frac{1}{2\pi R_1 C_1} = \frac{1}{2\pi(1.2 \times 10^3)(0.02 \times 10^{-6})} = \mathbf{6.63\ kHz}$$

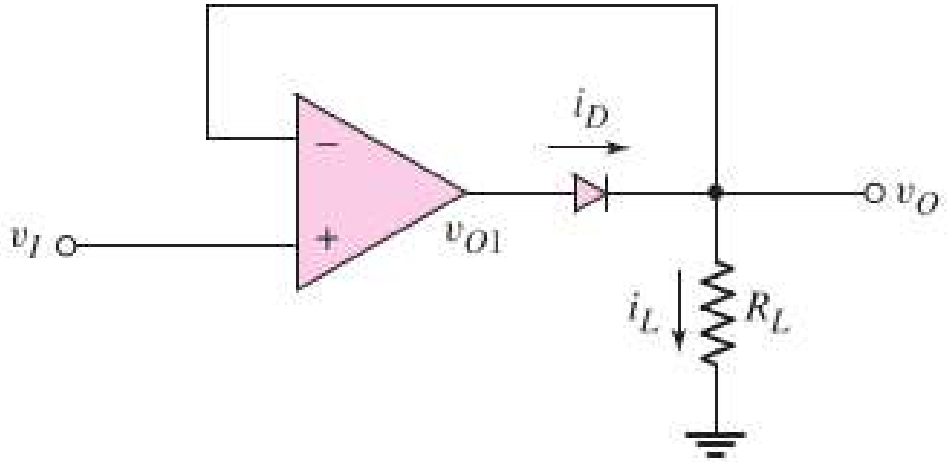
HPF

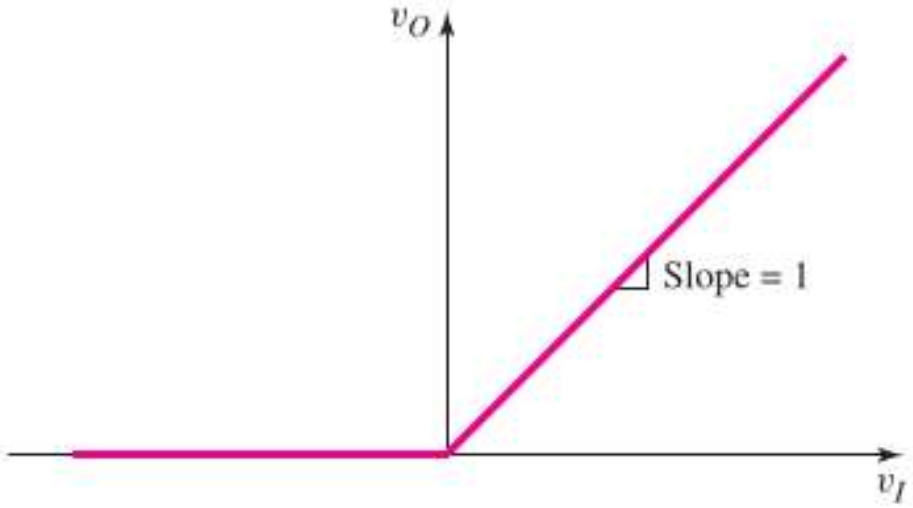


High-pass active filter cutoff frequency:

$$f_{OL} = \frac{1}{2\pi R_1 C_1}$$

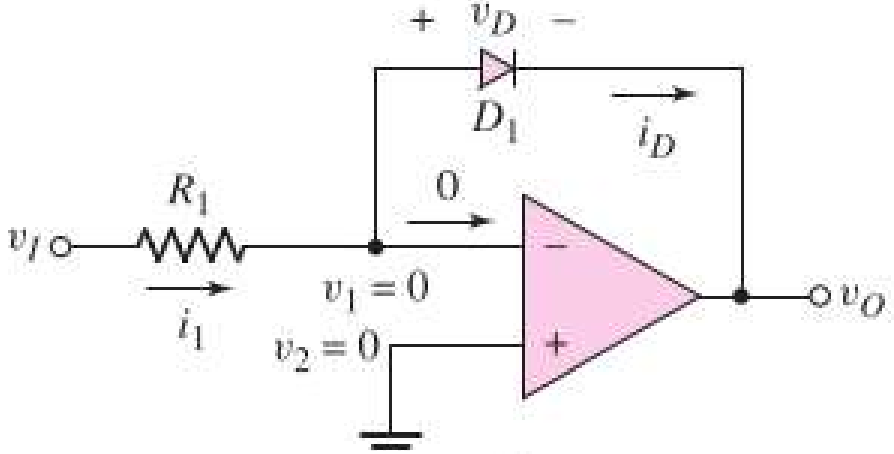
Nonlinear Circuit Applications





Half Wave Rectifier?

Log Amplifier



$$i_D = I_S(e^{v_D/V_T} - 1)$$

If the diode is sufficiently forward biased, the (-1) term is negligible, and

$$i_D \cong I_S e^{v_D/V_T}$$

The input current can be written

$$i_1 = \frac{v_I}{R_1}$$

and the output voltage, since v_1 is at virtual ground, is given by

$$v_O = -v_D$$

$$i_D = I_S(e^{v_D/V_T} - 1)$$

If the diode is sufficiently forward biased, the (-1) term is negligible, and

$$i_D \cong I_S e^{v_D/V_T}$$

The input current can be written

$$i_1 = \frac{v_I}{R_1}$$

and the output voltage, since v_1 is at virtual ground, is given by

$$v_O = -v_D$$

Noting that $i_1 = i_D$, we can write

$$i_1 = \frac{v_I}{R_1} = i_D = I_S e^{-v_O/V_T}$$

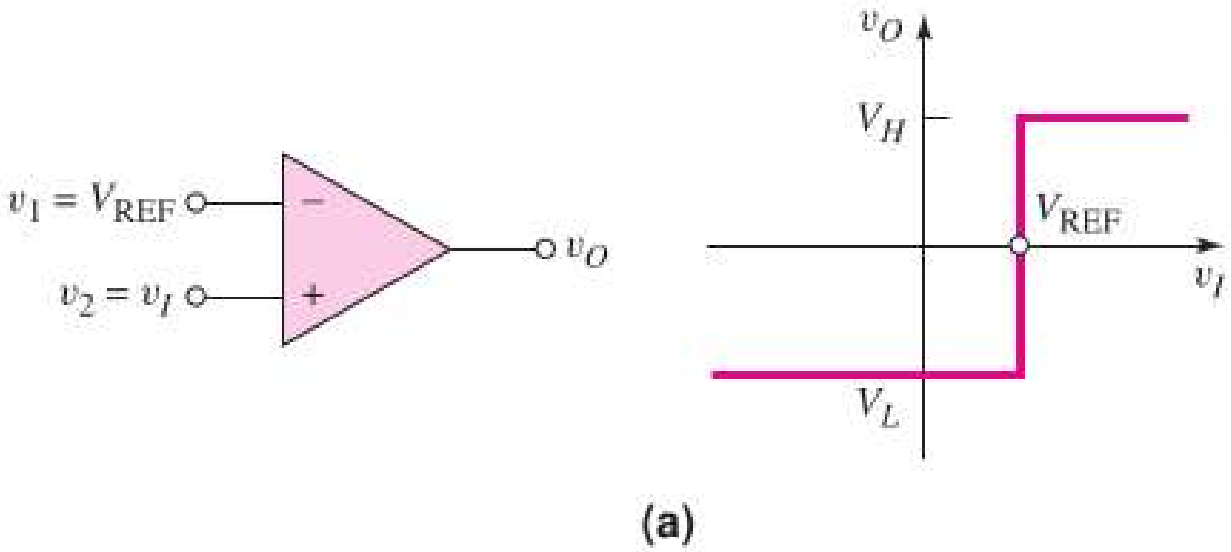
If we take the natural log of both sides of this equation, we obtain

$$\ln\left(\frac{v_I}{I_S R_1}\right) = -\frac{v_O}{V_T}$$

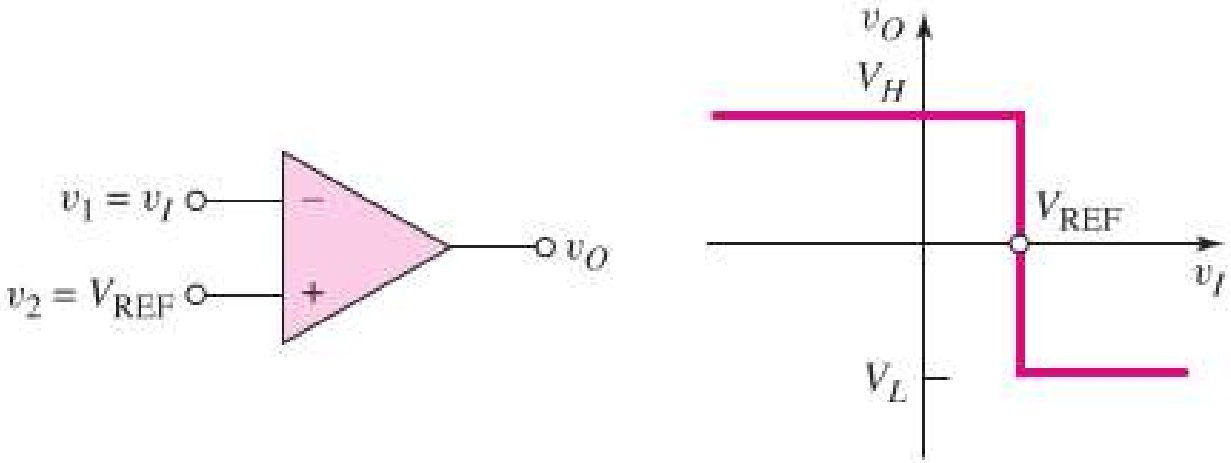
or

$$v_O = -V_T \ln\left(\frac{v_I}{I_S R_1}\right)$$

Comparator



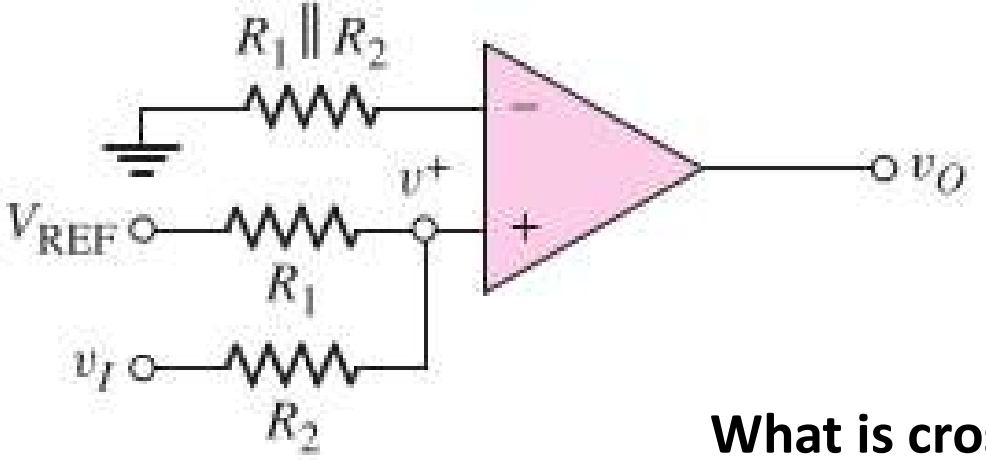
(a)



(b)

Figure 15.26 (a) Noninverting comparator circuit and (b) inverting comparator circuit

Comparator



What is value of V+ ?

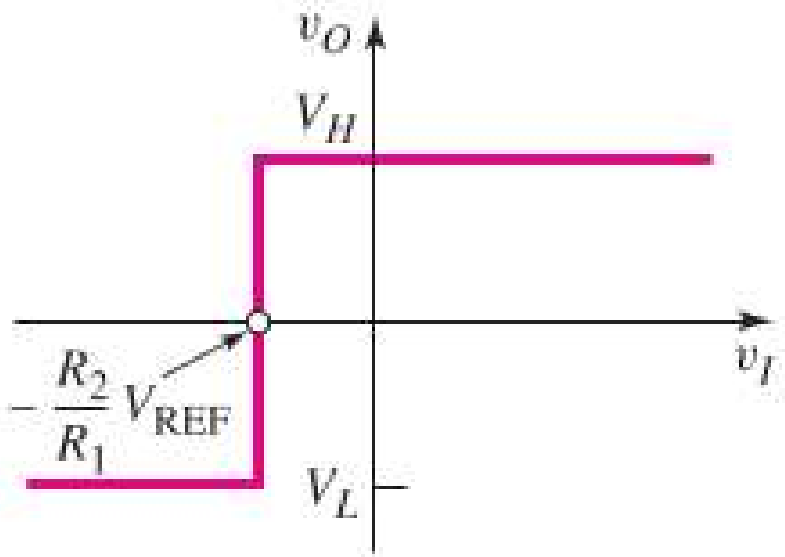
$$v_+ = \left(\frac{R_2}{R_1 + R_2}\right) V_{\text{REF}} + \left(\frac{R_1}{R_1 + R_2}\right) v_I$$

What is crossover voltage?
Ideally when V+ = 0 V

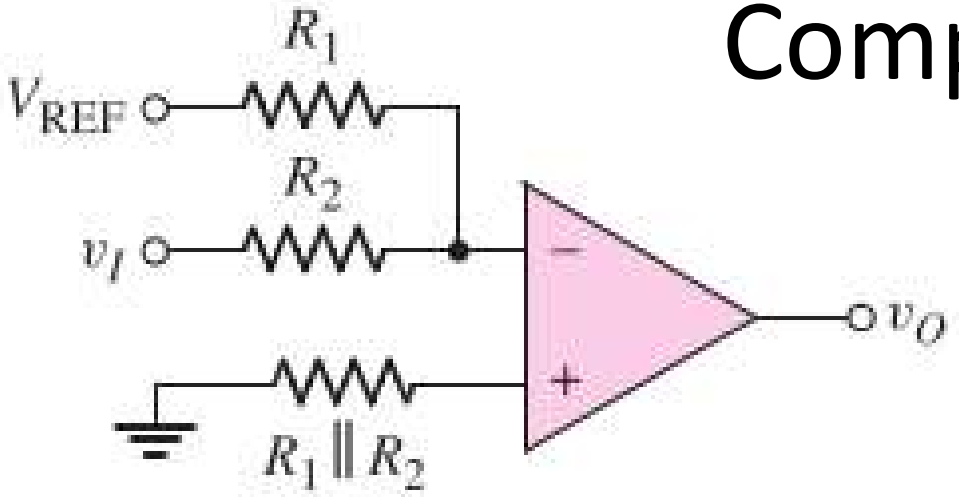
$$R_2 V_{\text{REF}} + R_1 v_I = 0$$

Draw Transfer characteristics of this circuit

$$v_I = -\frac{R_2}{R_1} V_{\text{REF}}$$



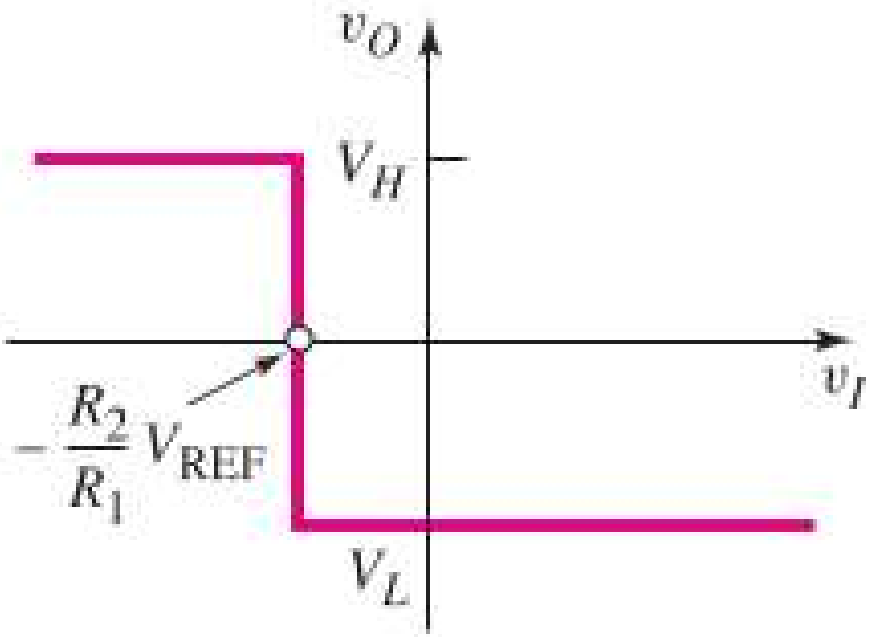
Comparator

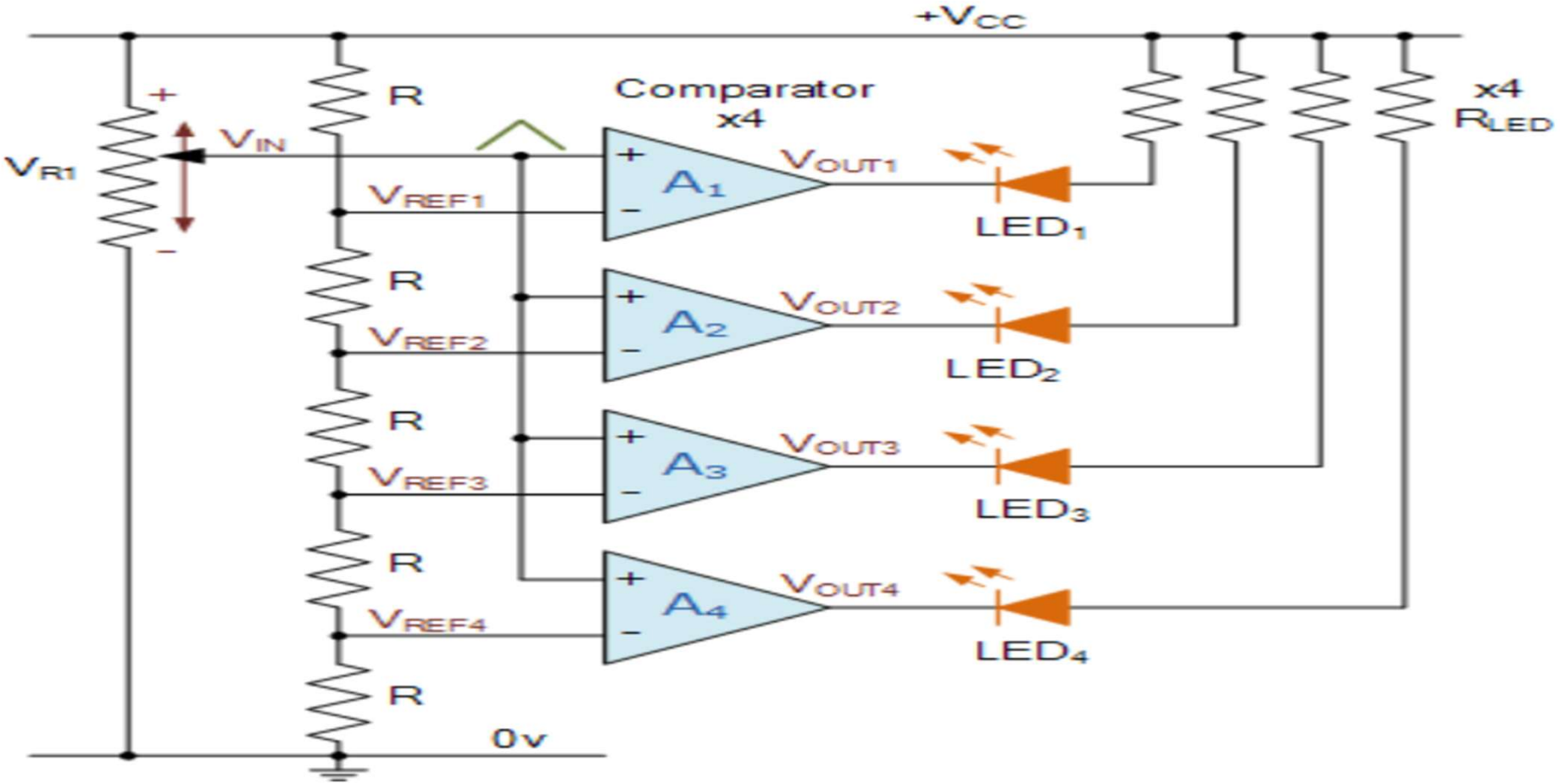


What is value of V- ?

What is crossover voltage?
Ideally when V- = 0 V

Draw Transfer characteristics of this circuit

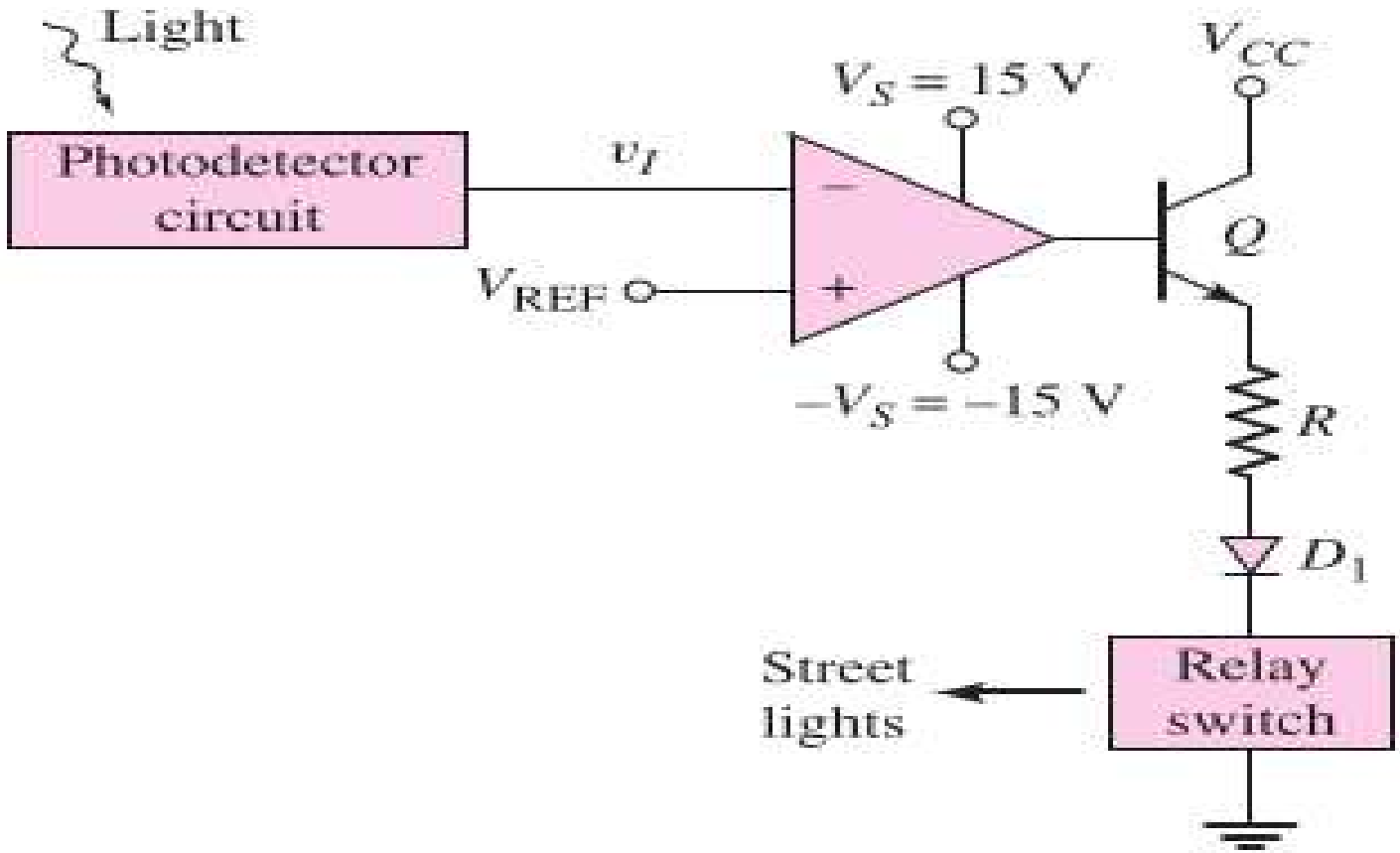




As the common input voltage increases, the output of each op-amp comparator circuit switches in turn thereby turning OFF the connected LED starting with the lower comparator, A_4 and upwards towards A_1 as the input voltage increases. So by setting the values of the resistors in the voltage divider network, the comparators can be configured to detect any voltage level. One good example of the use of voltage level detection and indication would be for a battery condition monitor by reversing the LED's and connecting them to 0V (ground) instead of V_{CC}.

Application of comparator

- To control Street Lights
- Input signal is output of photodetector circuit
- During night, tube lights should be ‘ON’
- During daytime, tube lights should be ‘OFF’



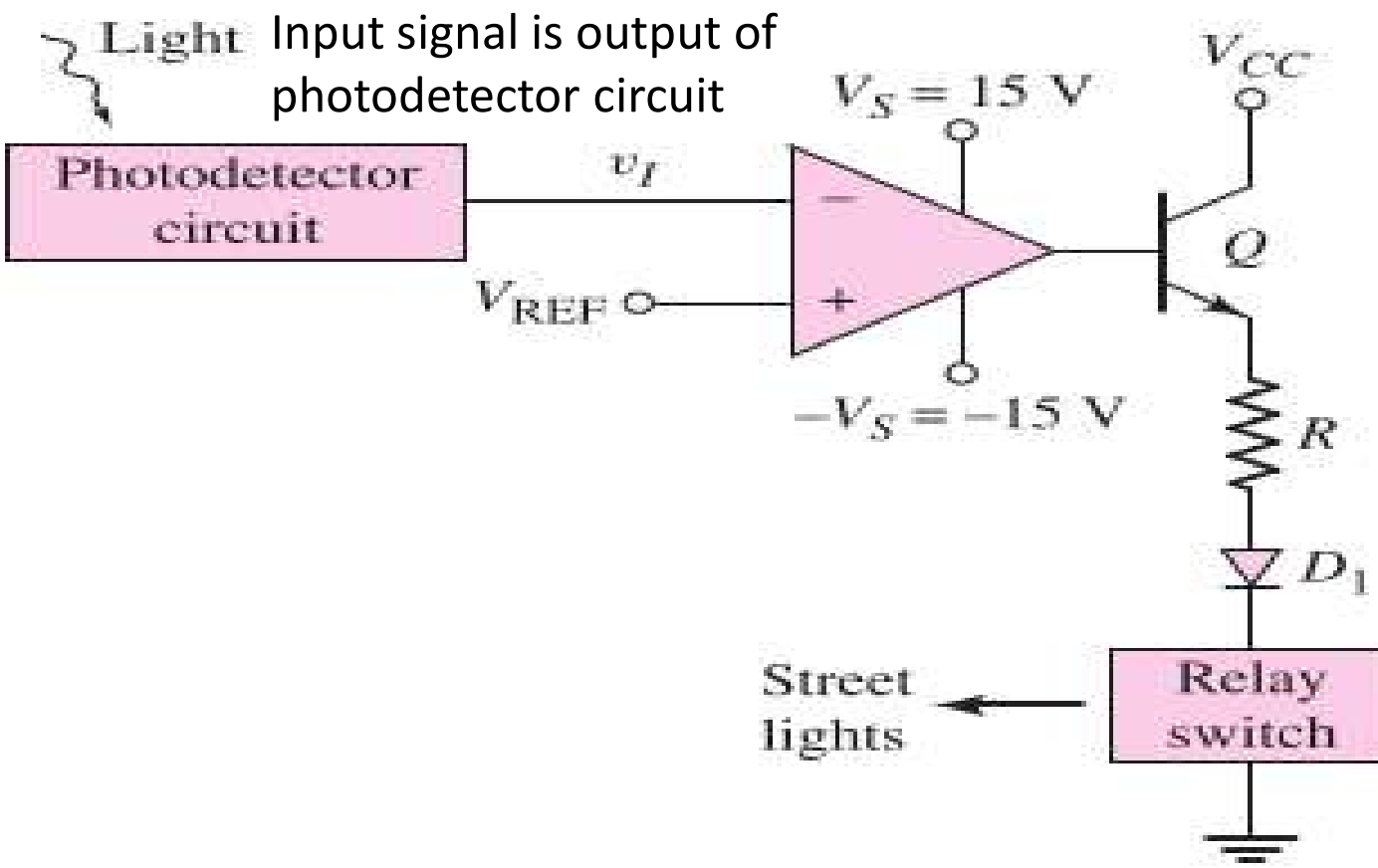
What is relay?

https://www.youtube.com/watch?v=cunddFiQzrk

https://www.youtube.com/watch?v=1lbdwPfFegY

https://www.youtube.com/watch?v=J0vi5TqqCyw

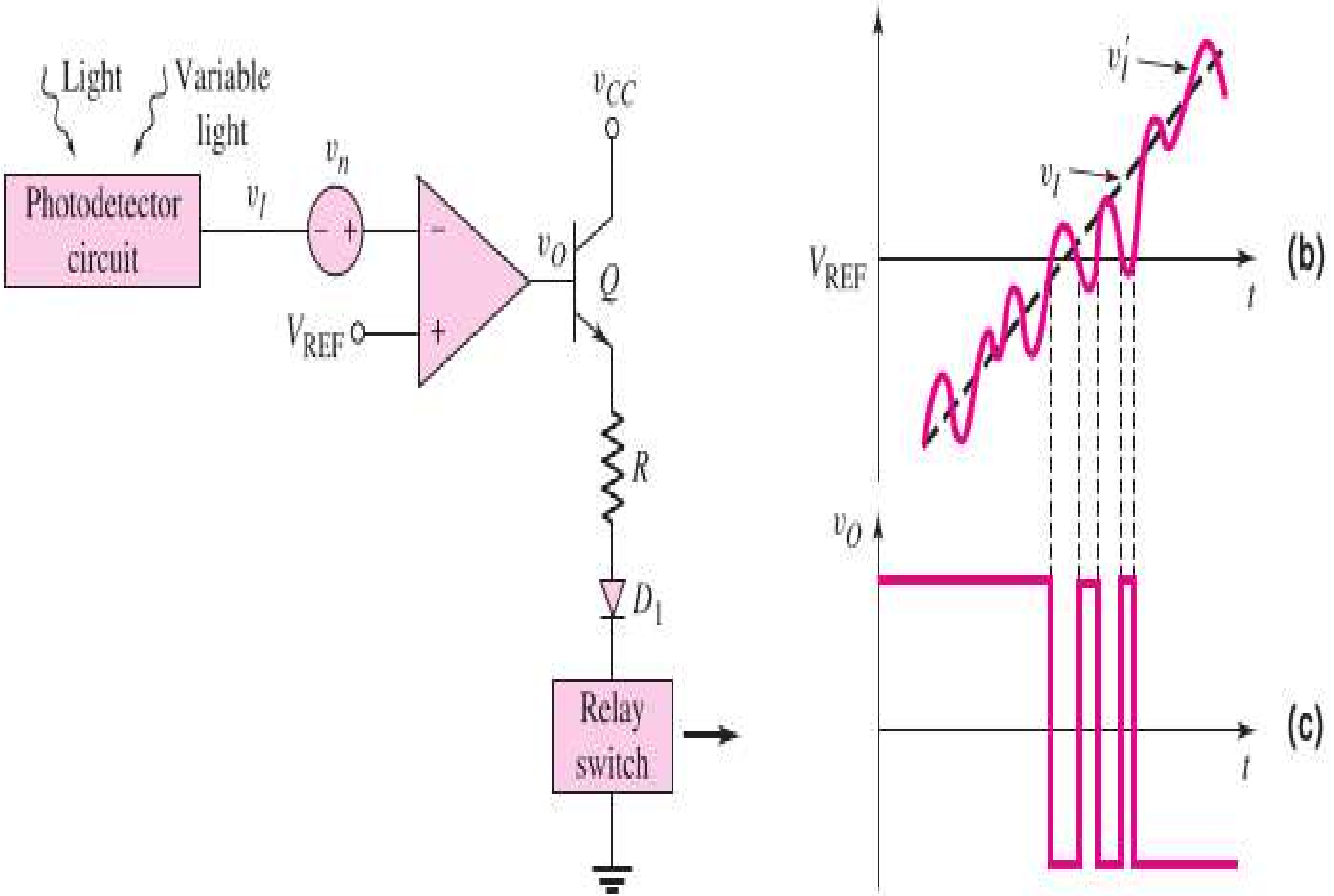
How does relay work?



During night, V_I < Vref
Vo = Vsat=15V
Transistor turns 'ON'
Current in relay , turns street lights ON

During Day, V_I > Vref
Vo = -Vsat= -15V
Transistor turns 'OFF'
Current in relay , turns street lights OFF

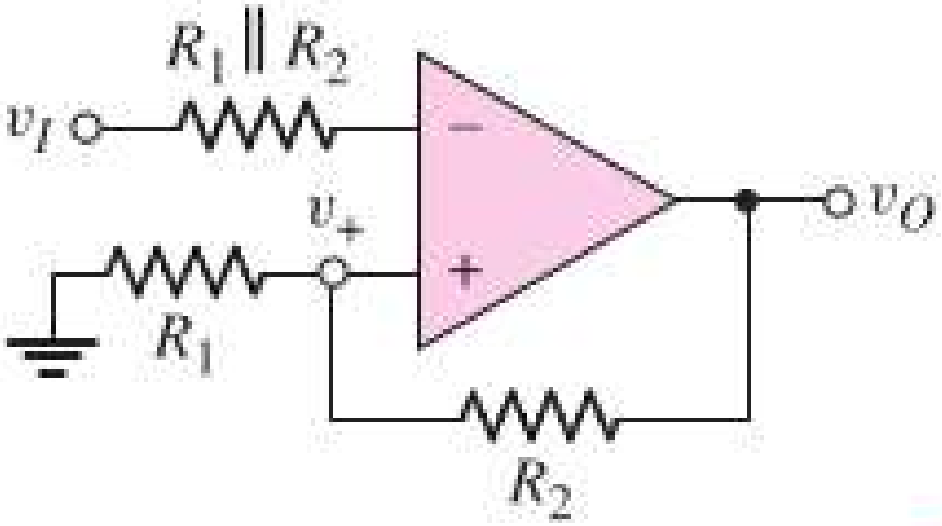
Diode D1..protection device


Light
Variable light
Photodetector circuit
v_l
v_n
V_{REF}
v_{CC}
v_O
Q
R
D_1
Relay switch
v_l'
v_l
V_{REF}
t
(b)
v_O
t
(c)

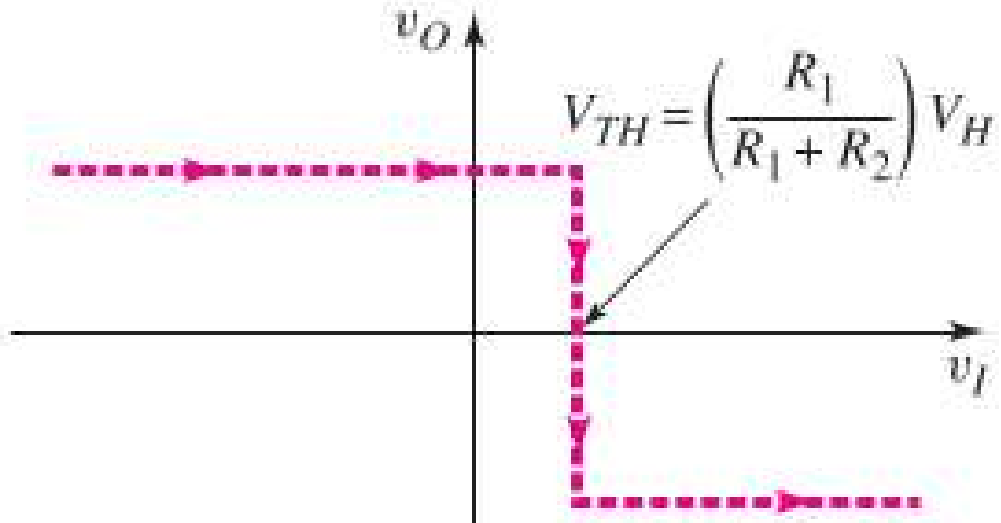
Schmitt Trigger

- Bistable Multivibrator
- Positive feedback, loop gain more than unity

V+ ??????

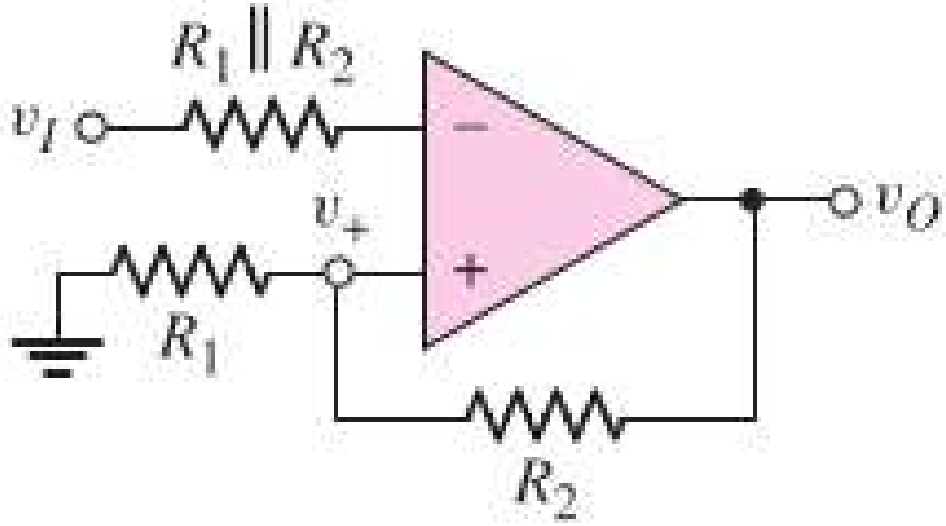


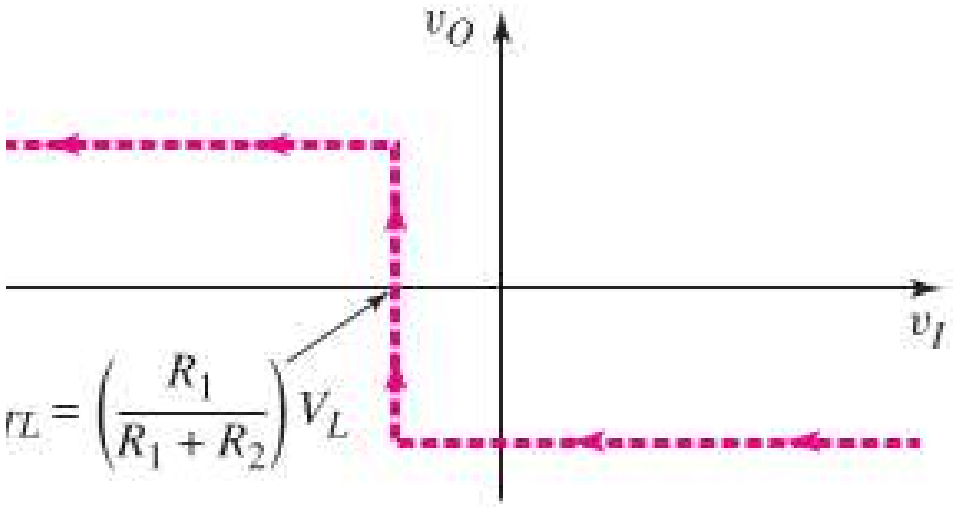
$$v_+ = \left(\frac{R_1}{R_1 + R_2}\right) v_O$$

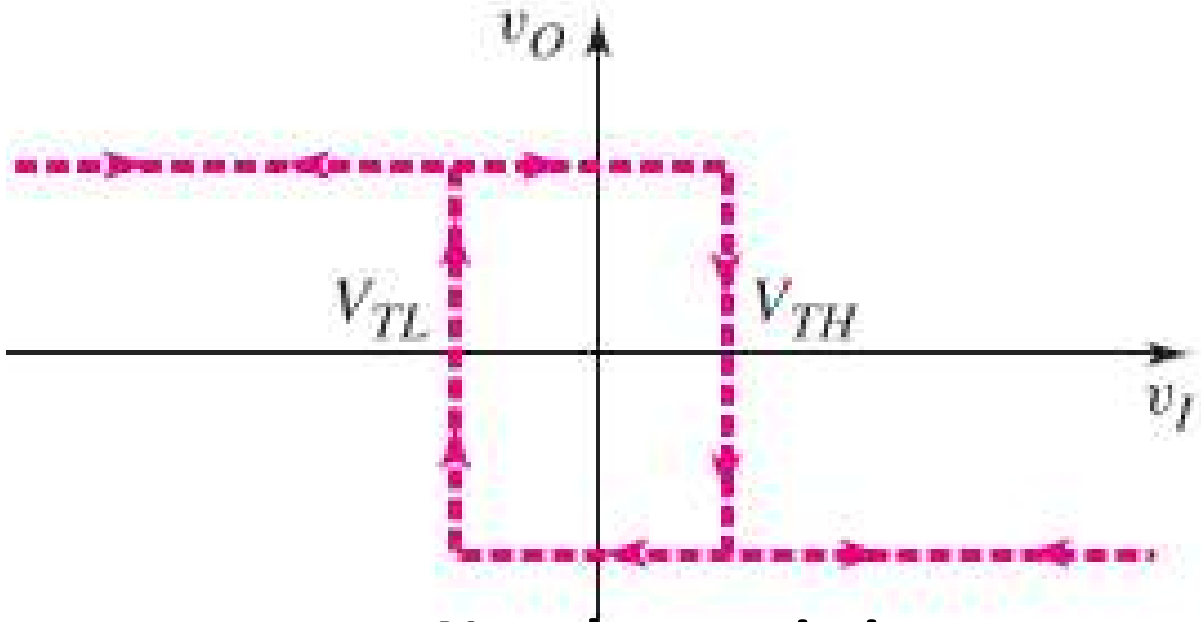


As input voltage increases

As input voltage decreases

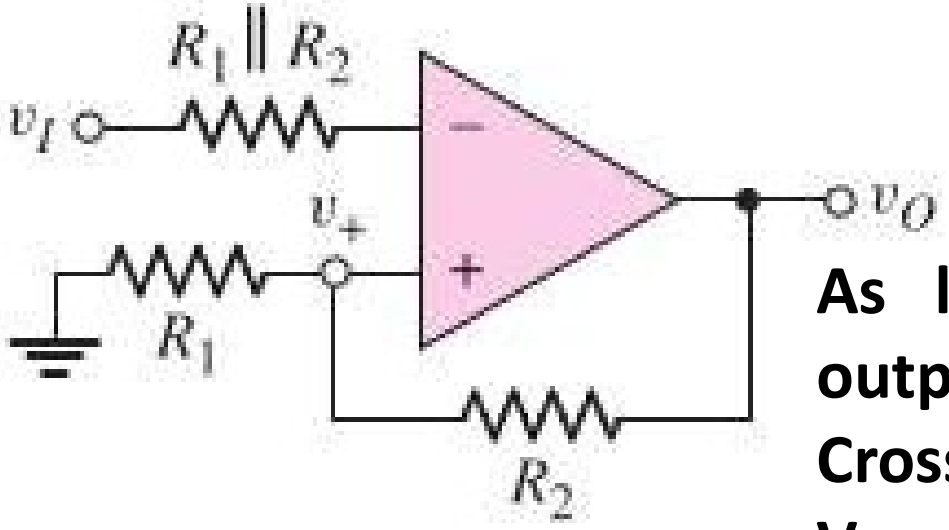






Net charactristics

Assume output of comparator is in one state Vo = V_H



$$v_+ = \left(\frac{R_1}{R_1 + R_2}\right) V_H$$

As long as input signal is less than V+, output remains in high state Vo = V_H

Crossover Voltage occurs when V_I = V+……..call as V_{TH}

$$V_{TH} = \left(\frac{R_1}{R_1 + R_2}\right) V_H$$

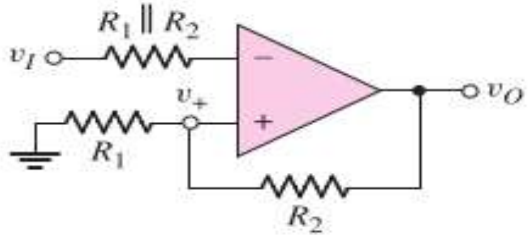
When when $V_I > V_{TH}$, Voltage at inverting terminal is greater than that at noninverting terminal

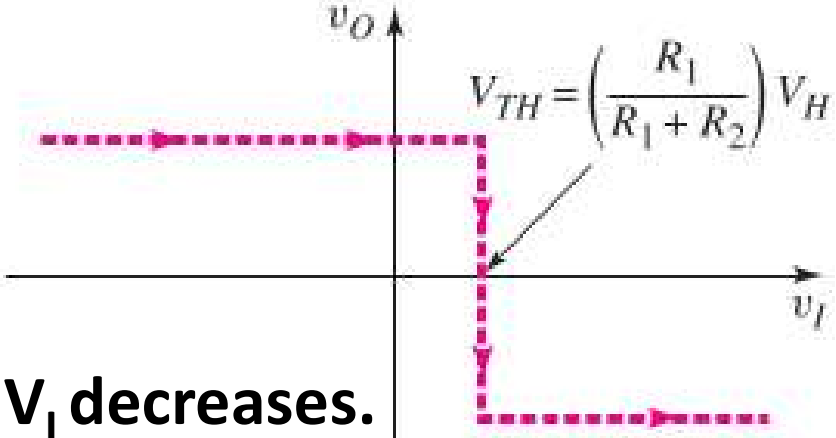
Differential input voltage ($V_I - V_{TH}$) is amplified by open loop comparator. And output switches to low state V_L

$$v_+ = \left(\frac{R_1}{R_1 + R_2}\right) V_L$$

When $V_L < V_H$, input Voltage V_I is still greater than V+, and output remains at low state as V_I continues to increase.

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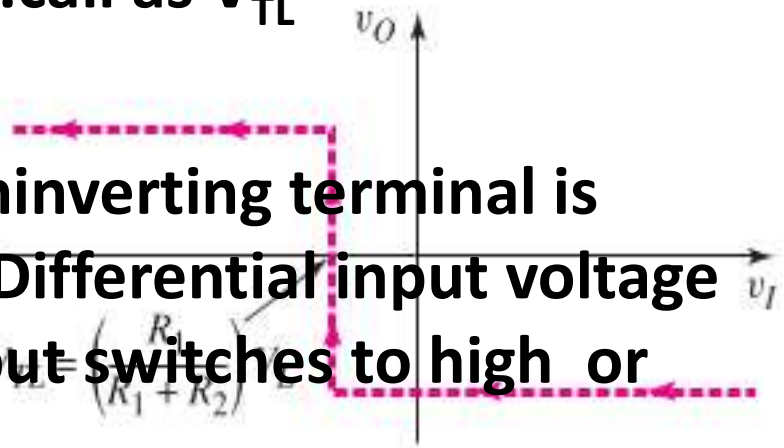


Now consider transfer characteristics as V_I decreases.

As long as V_I is greater than V+, V+ = (R1/(R1+R2)) V_L, output remains at low state

Crossover Voltage occurs when V_I = V+…….call as V_{TL}

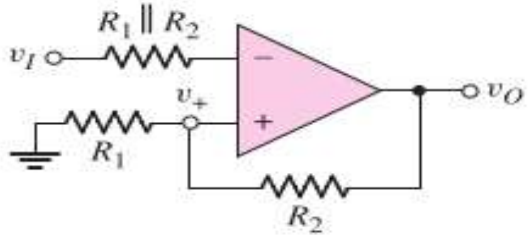
$V_{TL} = \left(\frac{R1}{R1+R2}\right)$

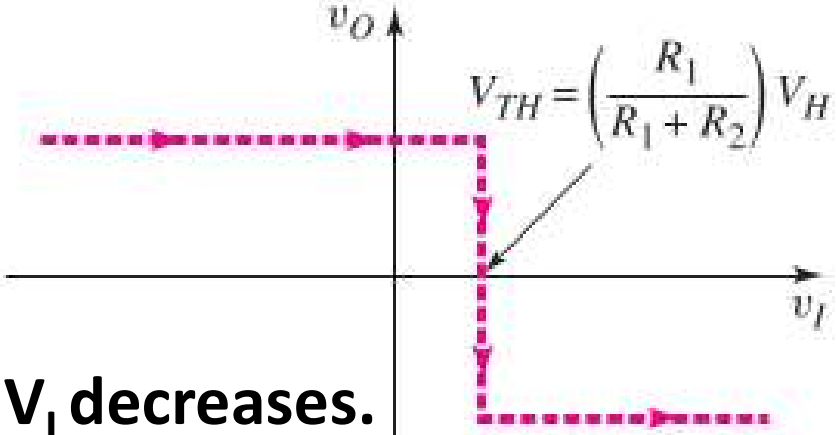


As V_I drops below this value, voltage at noninverting terminal is greater than voltage at inverting terminal. Differential input voltage is amplified by open loop comparator, output switches to high or $V_O = V_H$,

As V_I continues to decrease, it remains less than V+;
output remains at High state.

When $V_L < V_H$, input Voltage V_I is still greater than V+, and output remains at low state as V_I continues to increase.





Now consider transfer characteristics as V_I decreases.

As long as V_I is greater than V+, V+ = (R1/(R1+R2)) V_L, output remains at low state

Crossover Voltage occurs when V_I = V+.....

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